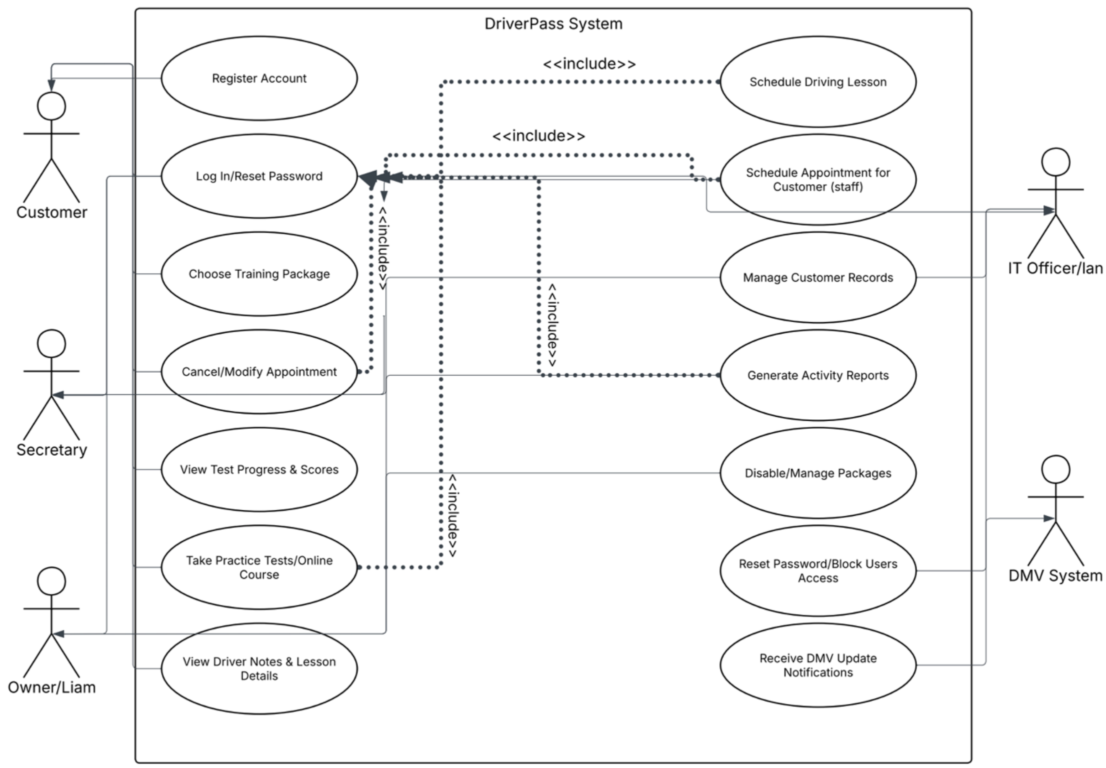
Southern New Hampshire University

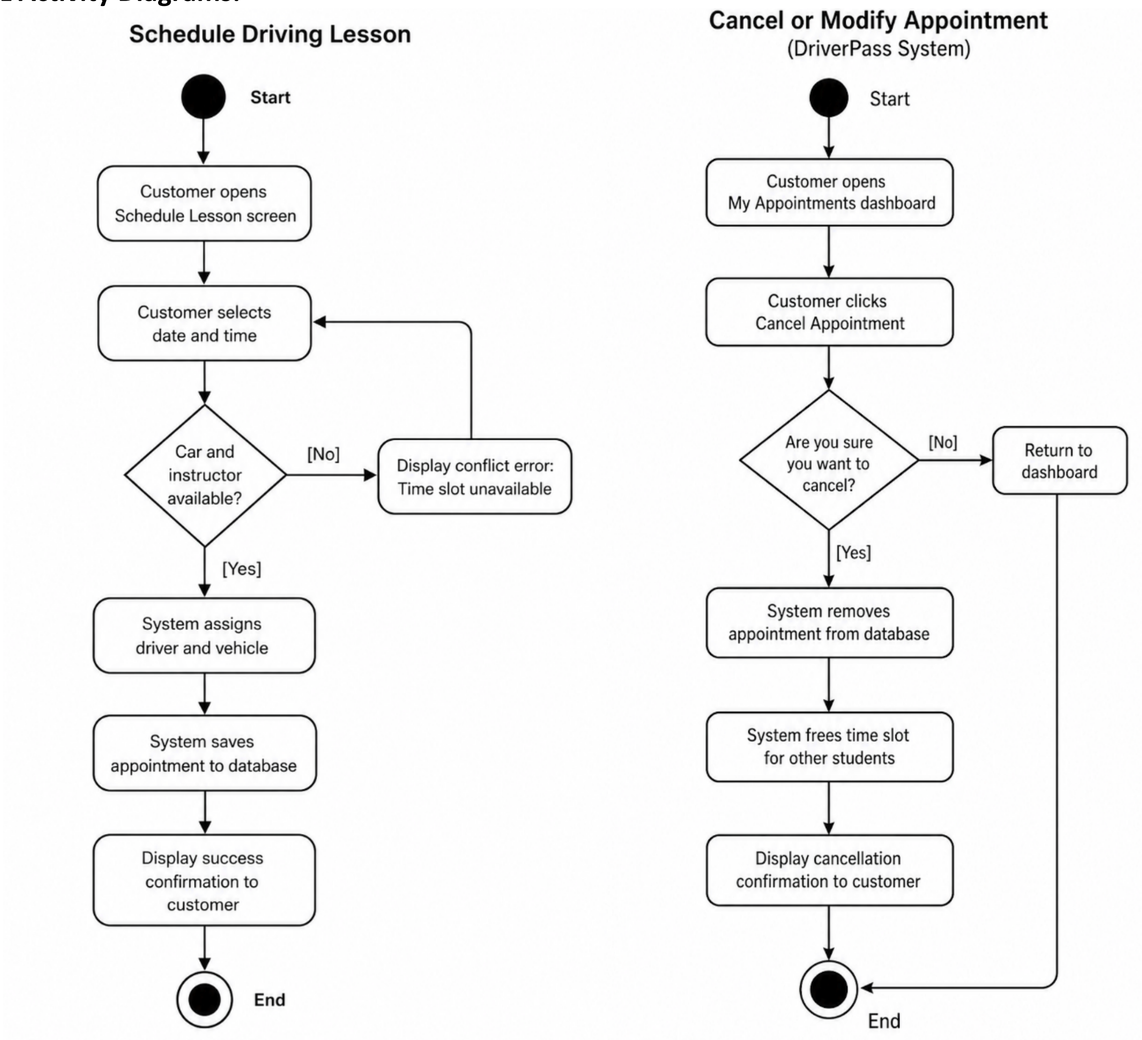
# CS 255 System Design

# UML Diagrams

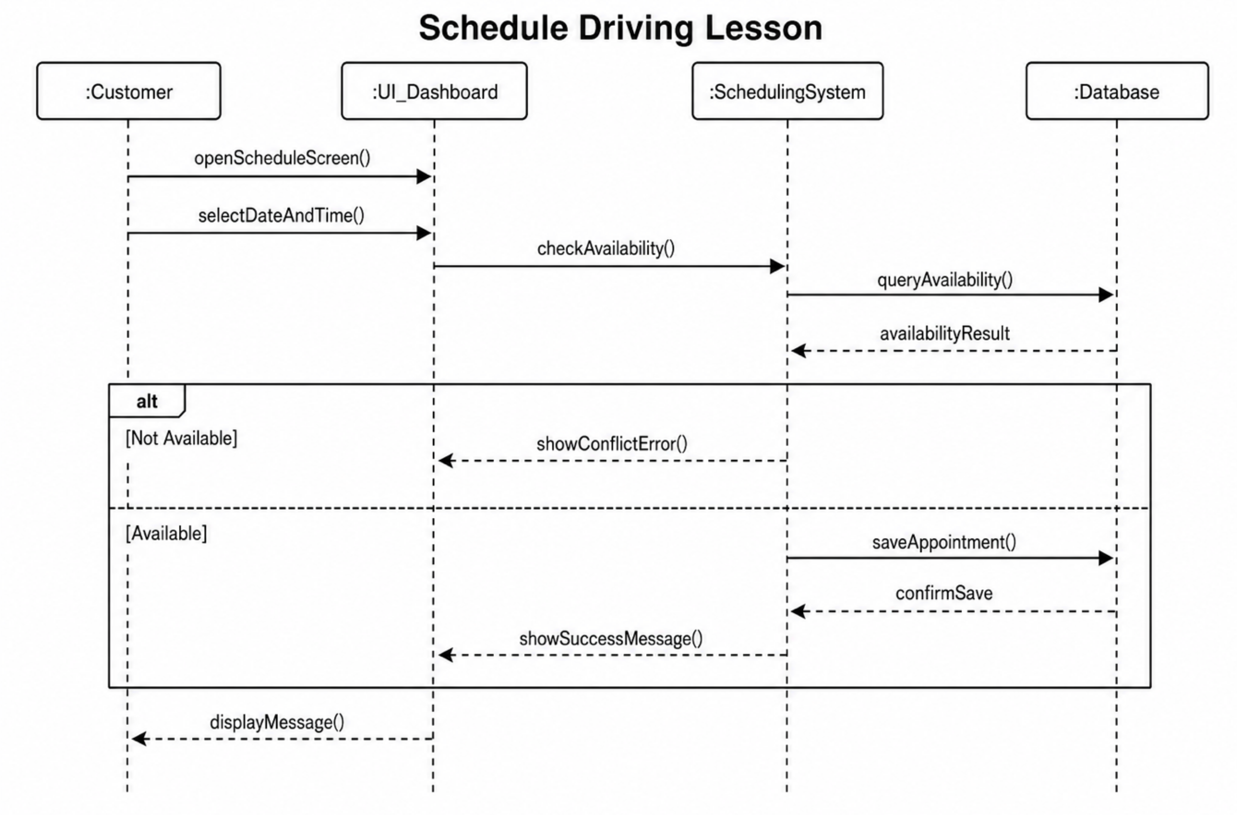
## UML Use Case Diagram:



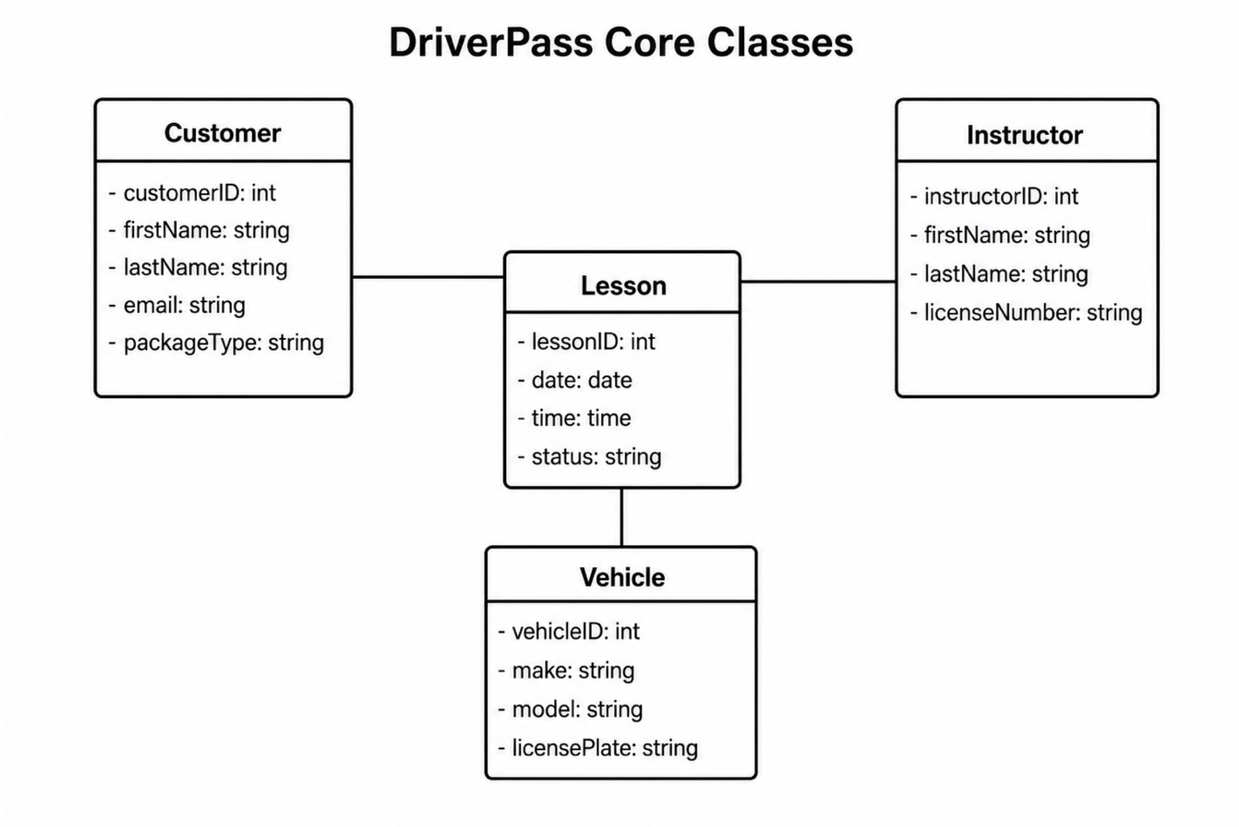
**UML Activity Diagrams:**



**UML Sequence Diagram:**



**UML Class Diagram:**




## Technical Requirements

For the technical requirements, the system needs to run in the cloud since DriverPass does not want to deal with managing their own servers or backups. It has to be accessible from any normal web browser on computers, tablets, and phones so customers do not have to download any special software to use it. Behind the scenes, the application will need a relational database to store and connect the customer, instructor, vehicle and lesson information that I laid out in the class diagram.

To keep everything secure and address the feedback from project one, the system relies heavily on role-based access. This means the UI will display different dashboards depending on who logs in. A regular customer will only see their own profile and scheduling options, while the IT officer gets the admin control needed to reset passwords and lock out accounts. The system also needs to be fast enough to load pages in under two seconds, so users do not get frustrated when trying to book a lesson. Finally, the code needs to include automatic checks and confirmation popups to catch scheduling conflicts and stop double booking right when it happens.